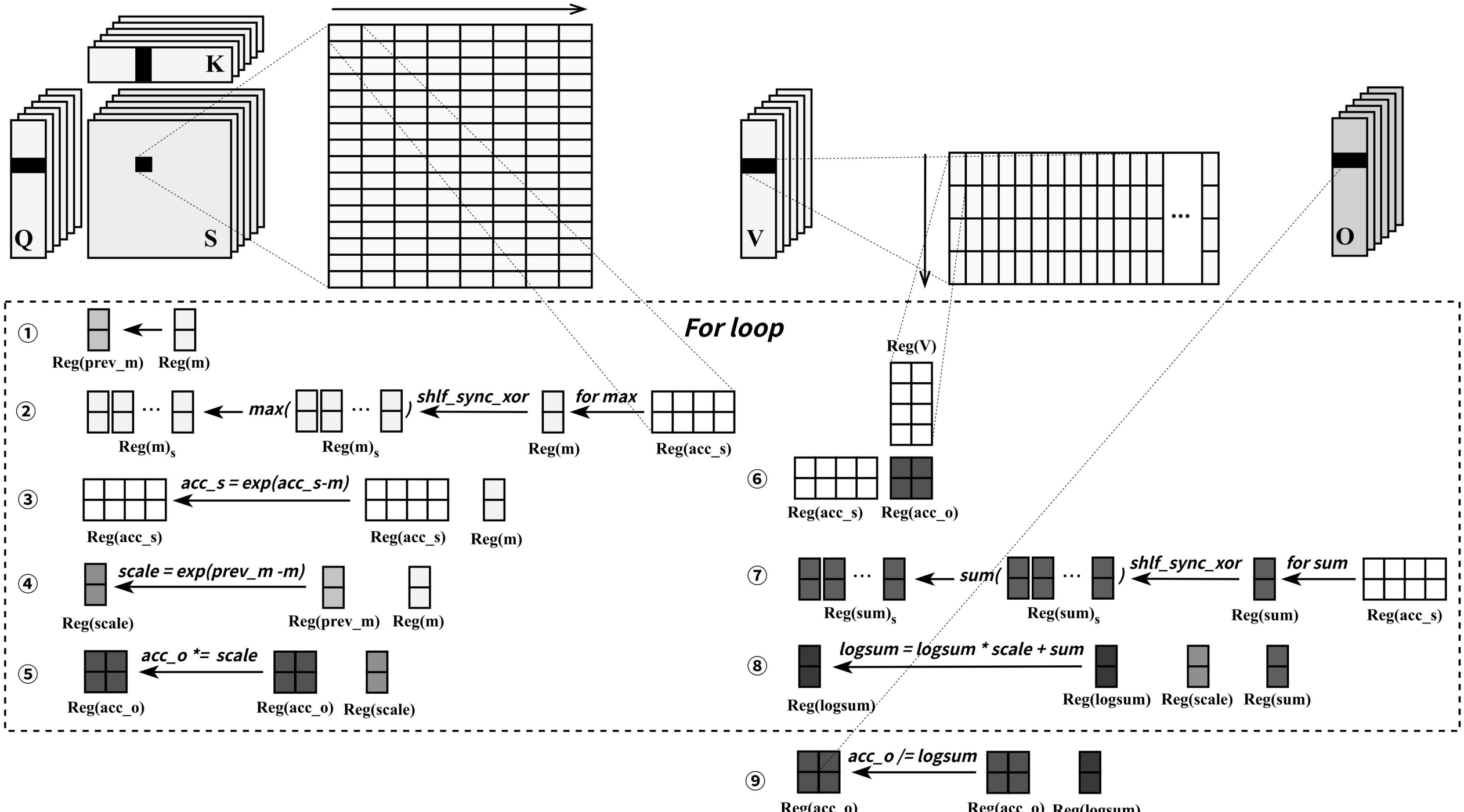
K
Q
S
V
O
...
For loop
Reg(V)
①
Reg(prev_m)
Reg(m)
②
max(
shlf_sync_xor
for max
Reg(m)s
Reg(m)s
Reg(m)
Reg(acc_s)
③
acc_s = exp(acc_s-m)
Reg(acc_s)
Reg(acc_s)
Reg(m)
④
scale = exp(prev_m -m)
Reg(scale)
Reg(prev_m)
Reg(m)
⑤
acc_o *= scale
Reg(acc_o)
Reg(acc_o)
Reg(scale)
⑥
Reg(acc_s)
Reg(acc_o)
⑦
sum(
shlf_sync_xor
for sum
Reg(sum)s
Reg(sum)s
Reg(sum)
Reg(acc_s)
⑧
logsum = logsum * scale + sum
Reg(logsum)
Reg(logsum)
Reg(scale)
Reg(sum)
⑨
acc_o /= logsum
Reg(acc_o)
Reg(acc_o)
Reg(logsum)